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END SEMESTER EXAMINATION 2024*Name of the Course:* B.Tech Petroleum Engineering*Semester:* VII*Name of the Course:* Advanced EOR Technique*Course Code:* TPE-717

Time: Three Hours

MM: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions in each main question.
- iii. Total marks for each main question is **twenty**
- iv. Each Question carries **10 marks**

Q1		COs																		
a.	Explain the classification of EOR as a thermal and non-thermal enhanced oil recovery?	CO1, CO2																		
b.	What is production phase? Explain the different oil production phase? Why we need EOR? Currently available EOR is based on which parameter?	CO1, CO2																		
c.	Explain the different factors to be consider during water flooding in the secondary oil production?	CO3																		
Q2																				
a.	Discuss the operational activities in pressure measurement during the production of crude oil?	CO3, CO4																		
b.	Write and define the different types oil field pressure calculation requirements Explain static and bottom hole pressure?	CO1, CO4																		
c.	The data of depth and pressure is given in the below<table><tr><td>Depth(m)</td><td>0</td><td>200</td><td>400</td><td>600</td><td>800</td><td>1000</td><td>1200</td><td>1400</td></tr><tr><td>Press.(atm)</td><td>51</td><td>55.3</td><td>59.5</td><td>63.6</td><td>81.2</td><td>99.0</td><td>119.0</td><td>141.0</td></tr></table> Calculate the following: - Pressure gradients. - Depth of gas- oil and gas water inter face - Pressures at GOC and WOC Given: Depth of top perforation = 1140m, GOC = 745m, WOC = 1160m	Depth(m)	0	200	400	600	800	1000	1200	1400	Press.(atm)	51	55.3	59.5	63.6	81.2	99.0	119.0	141.0	CO1, CO2
Depth(m)	0	200	400	600	800	1000	1200	1400												
Press.(atm)	51	55.3	59.5	63.6	81.2	99.0	119.0	141.0												
Q3																				
a.	Discuss primary reservoir mechanism in principle of water flooding? Discuss all the	CO1																		

	six primary mechanism in details?	
b.	Discuss the first and second Cole principle in water flooding? Discuss the different selection patterns of water flooding.	CO1. CO2
c.	Explain the displacement efficiency in terms of average water saturation, initial gas saturation and initial water saturation? Calculate: A saturated oil reservoir is under consideration to be waterflooded immediately after drilling and completion. Core analysis tests indicate that the initial and residual oil saturations are 70 and 35%, respectively. Calculate the displacement efficiency when the oil saturation is reduced to 65, 60, 55, 50, and 35%. Assume that B_o will remain constant throughout the project life.	CO2, CO5
Q4		
a.	With the help of ternary diagram explain all the diagram by considering the different pressure and temperature along with the phase diagram inside the ternary diagram	CO1. CO5
b.	What are the basic methods of miscible drive? Explain the phase conditions in the reservoir during high pressure gas injection.	CO4. CO5
c.	Explain the following: a. Miscible slug flooding b. Alcohol slug injection	CO3, CO6
Q5		
a.	Illustrate the Reservoir Selection and Reservoir Process design during advanced enhanced oil recovery.	CO4. CO6
b.	Define MEOR? Draw the Schematic diagram of MEOR and explain the each steps in increases the production rate of microorganisms during the enhanced oil recovery	CO4, CO5
c.	Explain Microbial Recovery Techniques? Discuss the different microbes identified to be used during MEOR. Explain MEOR Principles, Applications and Potential?	CO2. CO3